

Demonstration of Lattestex

In the 13 TeV era

Number 6, Number 1

Universidade de São Paulo

7 August 2026







Some centred text

Some more centred text

Some centred text

2026-08-07

7 August 2026

August 2026

- Item
- Item
 - Subitem
 - Subitem
 - Subitem
- Item
- Item

- Item

- Item
- Item

- Item
- Item
 - Subitem

- Item
- Item
 - Subitem
 - Subitem

- Item
- Item
 - Subitem
 - Subitem
 - Subitem

- Item
- Item
 - Subitem
 - Subitem
 - Subitem
- Item

- Item
- Item
 - Subitem
 - Subitem
 - Subitem
- Item
- Item

- 1 Item
- 2 Item
 - 1 Subitem
 - 2 Subitem
 - 1 Subitem
- 3 Item
- 4 Item

A Item

B Item

 A Subitem

 B Subitem

 A Subitem

C Item

D Item

- ✓ Item
- ✓ Item
 - ✓ Subitem
 - Subitem
 - Subitem
- ✓ Item
- ✗ Item

An attached ROOT data file:

↓ data.root

- URL: <http://info.cern.ch/hypertext/WWW/TheProject.html>
- Hyperlink: TheProject
- Hyperlink: ATL-COM-PHYS-2014-1471

- “quotation” ‘marks’

- $H^+ \rightarrow tb$
- Lepton p_T and η

Jet pull is a variable that can be constructed from particles within a jet cone. The procedure is as follows:

- Select a pair of jets in an event.
- Build a vector sum of calorimeter cells within each jet.

$$\vec{p} = \sum_i \frac{E_T^i |r_i|}{E_T^{\text{jet}}} \vec{r}_i$$

- $\vec{r}_i$: position of jet cell i relative to jet centre
- E_T^i : transverse energy of cell i
- E_T^{jet} : transverse energy of jet

Each cell is assigned to the closer jet in (η, ϕ) space.

Jet pull is a variable that can be constructed from particles within a jet cone. The procedure is as follows:

- Select a pair of jets in an event.

Jet pull is a variable that can be constructed from particles within a jet cone. The procedure is as follows:

- Select a pair of jets in an event.
- Build a vector sum of calorimeter cells within each jet.

Jet pull is a variable that can be constructed from particles within a jet cone. The procedure is as follows:

- Select a pair of jets in an event.
- Build a vector sum of calorimeter cells within each jet.

$$\vec{p} = \sum_i \frac{E_T^i |r_i|}{E_T^{\text{jet}}} \vec{r}_i$$

- $\vec{r}_i$: position of jet cell i relative to the jet centre

Jet pull is a variable that can be constructed from particles within a jet cone. The procedure is as follows:

- Select a pair of jets in an event.
- Build a vector sum of calorimeter cells within each jet.

$$\vec{p} = \sum_i \frac{E_T^i |r_i|}{E_T^{\text{jet}}} \vec{r}_i$$

- $\vec{r}_i$: position of jet cell i relative to the jet centre
- E_T^i : transverse energy of cell i

Jet pull is a variable that can be constructed from particles within a jet cone. The procedure is as follows:

- Select a pair of jets in an event.
- Build a vector sum of calorimeter cells within each jet.

$$\vec{p} = \sum_i \frac{E_T^i |r_i|}{E_T^{\text{jet}}} \vec{r}_i$$

- $\vec{r}_i$: position of jet cell i relative to the jet centre
- E_T^i : transverse energy of cell i
- E_T^{jet} : transverse energy of the jet

Jet pull is a variable that can be constructed from particles within a jet cone. The procedure is as follows:

- Select a pair of jets in an event.
- Build a vector sum of calorimeter cells within each jet.

$$\vec{p} = \sum_i \frac{E_T^i |r_i|}{E_T^{\text{jet}}} \vec{r}_i$$

- $\vec{r}_i$: position of jet cell i relative to the jet centre
- E_T^i : transverse energy of cell i
- E_T^{jet} : transverse energy of the jet

Jet pull is a variable that can be constructed from particles within a jet cone. The procedure is as follows:

- Select a pair of jets in an event.
- Build a vector sum of calorimeter cells within each jet.

$$\vec{p} = \sum_i \frac{E_T^i |r_i|}{E_T^{\text{jet}}} \vec{r}_i$$

- $\vec{r}_i$: position of jet cell i relative to the jet centre
- E_T^i : transverse energy of cell i
- E_T^{jet} : transverse energy of the jet

Each cell is assigned to the closer jet in (η, ϕ) space.

Python code inline is like this: `print("hello")`

Python as a code block is like this:

```
import multiprocessing

def function(x):
    return x * x

if __name__ == '__main__':
    pool = multiprocessing.Pool(processes=4)
    result = pool.apply_async(function, [10])
    print(result.get(timeout=1))
    print(pool.map(function, range(10)))
```

Python code inline is like this: `print("hello")`

Python as a code block is like this:

```
import multiprocessing
```

```
def function(x):  
    return x * x
```

```
if __name__ == '__main__':  
    pool = multiprocessing.Pool(processes=4)  
    result = pool.apply_async(function, [10])  
    print(result.get(timeout=1))  
    print(pool.map(function, range(10)))
```

```
import sys
import time

def is_prime(n):
    return zip([True, False], decompose(n))[-1][0]

class IsPrimeCached(dict):
    def __missing__(self, n):
        r = is_prime(n)
        self[n] = r
        return r

is_prime_cached = IsPrimeCached()

def primes():
    yield 2
    n = 3
    while n < sys.maxint - 2:
        yield n
        n += 2
        while n < sys.maxint - 2 and not is_prime_cached[n]:
            n += 2

def decompose(n):
    for p in primes():
        if p*p > n: break
        while n % p == 0:
            yield p
            n /= p
    if n > 1:
        yield n
```

```
#include <iostream>
#include <thread>

void say_hello();

int main(int argc, char** argv){

    std::thread my_first_thread(say_hello);
    my_first_thread.join();

    return 0;
}

void say_hello(){
    std::cout << "hello from a thread" << std::endl;
}
```

```
#include <iostream>
#include <thread>

void say_hello();

int main(int argc, char** argv){

    std::thread my_first_thread(say_hello);
    my_first_thread.join();

    return 0;
}

void say_hello(){
    std::cout << "hello from a thread" << std::endl;
}
```

```
\documentclass{article}
\title{Processes involving charged mesons}
\author{C. ~M. ~G. ~Lattes, H. ~Muirhead, G. ~P. ~S. ~Occhialini, C. ~F. ~Powell}
\begin{document}
\maketitle
```

In recent investigations with the photographic method, it has been shown that slow charged particles of small mass, present as a component of the cosmic radiation at high altitudes, can enter nuclei and produce disintegrations with the emission of heavy particles. It is convenient to apply the term 'meson' to any particle with a mass intermediate between that of a proton and an electron.

```
\end{document}
```

```
\documentclass{article}
\title{Processes involving charged mesons}
\author{C. ~M. ~G. ~Lattes, H. ~Muirhead, G. ~P. ~S. ~Occhialini,
↪ C. ~F. ~Powell}
\begin{document}
\maketitle
```

In recent investigations with the photographic method, it has been

- ↪ shown that slow charged particles of small mass, present as a
- ↪ component of the cosmic radiation at high altitudes, can enter
- ↪ nuclei and produce disintegrations with the emission of heavy
- ↪ particles. It is convenient to apply the term `meson` to any
- ↪ particle with a mass intermediate between that of a proton and
- ↪ an electron.

```
\end{document}
```

Block 1

- Item
- Item

Block 2

- Item
- Item

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean commodo ligula eget dolor. Aenean massa. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec quam felis, ultricies nec, pellentesque eu, pretium quis, sem.

Nulla consequat massa quis enim. Donec pede justo, fringilla vel, aliquet nec, vulputate eget, arcu. In enim justo, rhoncus ut, imperdiet a, venenatis vitae, justo. Nullam dictum felis eu pede mollis pretium.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean commodo ligula eget dolor. Aenean massa. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec quam felis, ultricies nec, pellentesque eu,

pretium quis, sem. Nulla consequat massa quis enim. Donec pede justo, fringilla vel, aliquet nec, vulputate eget, arcu. In enim justo, rhoncus ut, imperdiet a, venenatis vitae, justo. Nullam dictum felis eu pede mollis pretium.

Lorem ipsum
dolor sit amet,
consectetur
adipiscing elit.
Aenean com-
modo ligula eget
dolor. Aenean
massa. Cum

sociis natoque
penatibus et mag-
nis dis parturi-
ent montes, nas-
cetur ridiculus
mus. Donec
quam felis, ul-
trices nec, pel-

lentesque eu, pre-
tium quis, sem.
Nulla consequat
massa quis enim.
Donec pede justo,
fringilla vel, ali-
quet nec, vulpu-
tate eget, arcu. In

enim justo, rhon-
cus ut, imperdiet
a, venenatis vitae,
justo. Nullam
dictum felis eu
pede mollis pre-
tium.

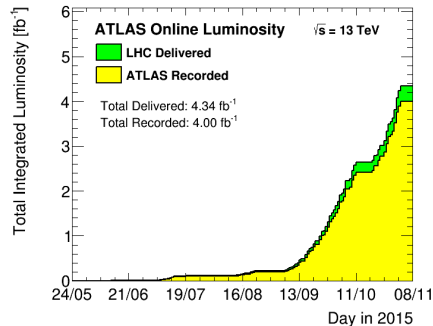
- Item 1
- Item 2
- Item 3

- Item 1
- Item 2
- Item 3

- Item 1
- Item 2
- Item 3

- Item 1
- Item 2
- Item 3

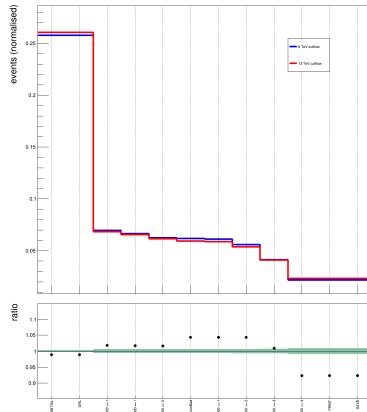
- Suppressed with respect to other Higgs modes
- $H \rightarrow b\bar{b}$ has the largest branching ratio (0.577 for m_H 125 GeV)
- Irreducible background from $t\bar{t}b\bar{b}$
- Other backgrounds: $t\bar{t}$ production in association with light quarks (u, d, s) or gluon jets (called $t\bar{t} + \text{light}$), and $t\bar{t} + c\bar{c}$



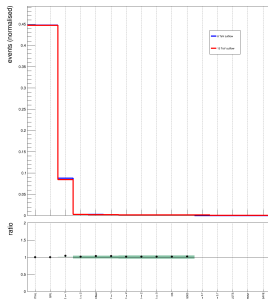
$\sqrt{s}$ (TeV)	7	8	13	14
$t\bar{t}H$ ($m_H = 125 \text{ GeV}$) (pb)	0.086	0.130	0.5085	0.611
$t\bar{t}$ (pb)	177	253	832	950
$S/\sqrt{B}$	0.00646	0.0082	0.0176	0.0198

7 TeV $\rightarrow$ 13/14 TeV: $S/\sqrt{B}$ changes by factor of $\simeq 3$

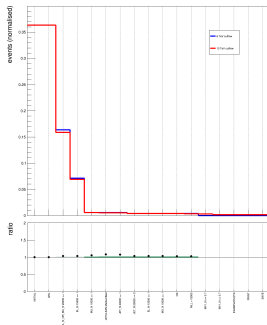
8 TeV cutflow versus 13 TeV cutflow for selection mu_jets



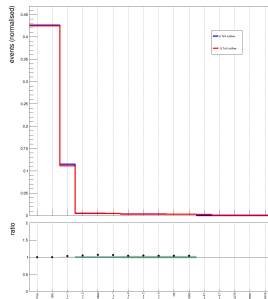
8 TeV cutflow versus 13 TeV cutflow for selection ee



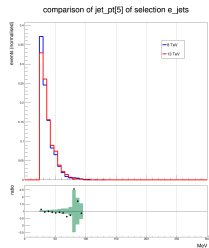
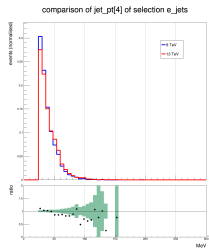
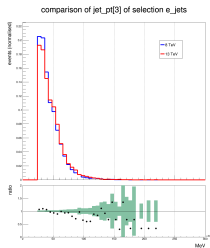
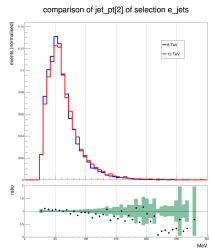
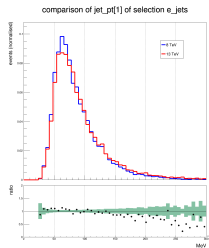
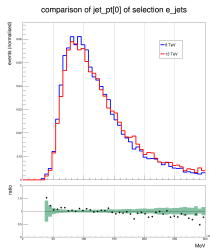
8 TeV cutflow versus 13 TeV cutflow for selection emu



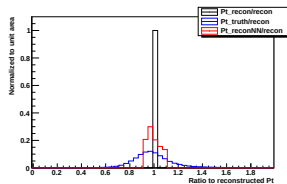
8 TeV cutflow versus 13 TeV cutflow for selection mumu



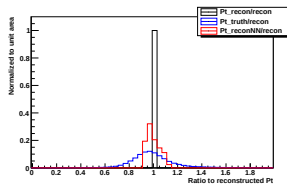
I3PD+SV1: <https://indico.cern.ch/event/387410/contribution/9/material/slides/0.pdf>



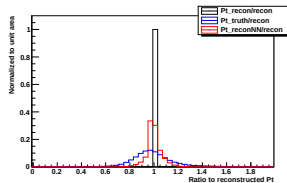
40 epochs:



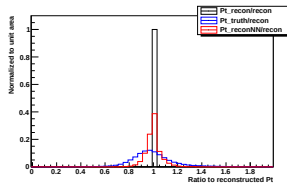
100 epochs:



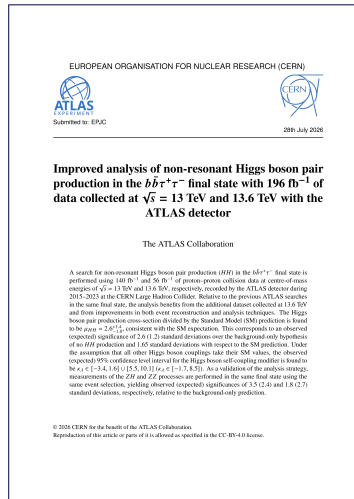
145 epochs:



300 epochs:



- Analysis paper (full Run-2 140 fb^{-1} at $\sqrt{s} = 13 \text{ TeV}$ & partial Run-3 56 fb^{-1} at $\sqrt{s} = 13.6 \text{ TeV}$) approved for ICHEP (final signoff 28 July)
- <https://gitlab.cern.ch/atlas-physics-office/HIGP/ANA-HIGP-2024-37/ANA-HIGP-2024-37-PAPER>
- <https://cds.cern.ch/record/2967281/files/HIGP-2024-37-002.pdf>
- Supporting internal note: <https://cds.cern.ch/record/2943884>
- Main ICHEP presentation: <https://indico.cern.ch/event/1522800/contributions/6987254>
- Preparations underway for full Run-3 analysis: <https://indico.cern.ch/event/1713662>



$M_{b\bar{b}}$ resolutions for $VH_{b\bar{b}}$ for progressively decreasing MET energy cut requirements for various neural networks, shown to 3 significant figures:

Selection	Events	NN0	NN1	NN2	NN3
$VH_{b\bar{b}}$	23686	0.133	0.129	0.131	0.131
$VH_{b\bar{b}} + \text{MET} < 100 \text{ GeV}$	22654	0.132	0.130	0.129	0.131
$VH_{b\bar{b}} + \text{MET} < 70 \text{ GeV}$	21094	0.131	0.128	0.129	0.129
$VH_{b\bar{b}} + \text{MET} < 40 \text{ GeV}$	15050	0.128	0.126	0.126	0.126
$VH_{b\bar{b}} + \text{MET} < 20 \text{ GeV}$	6174	0.130	0.127	0.126	0.127

Here, the physical processes are ranked according to the effectiveness of the corresponding behaviour they induce in NN3, where a greater effectiveness is taken to mean a smaller resolution value. *Caveat:* Systematic uncertainties are not given their due consideration.

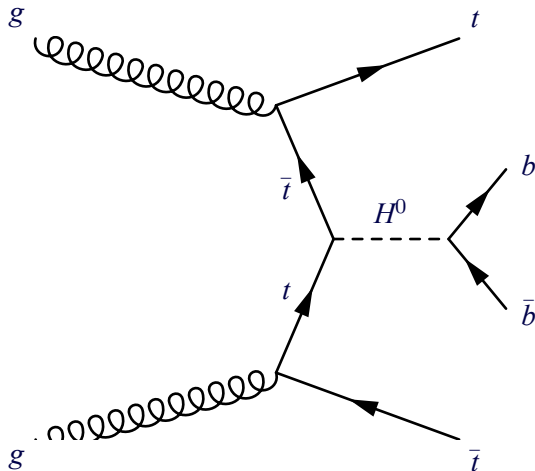
Selection	Events	NN0	NN1	NN2	NN3
$VH_{b\bar{b}} + \text{MET} > 100 \text{ GeV}$	1032	0.121542	0.129038	0.13072	0.116975
$VH_{b\bar{b}} + \text{MET} < 40 \text{ GeV}$	15050	0.128387	0.125939	0.125637	0.125963
$VH_{b\bar{b}} + \text{MET} < 20 \text{ GeV}$	6174	0.129539	0.127454	0.126029	0.127043
$VH_{b\bar{b}} + \text{MET} < 70 \text{ GeV}$	21094	0.131248	0.128119	0.128908	0.128825
$VH_{b\bar{b}} + \text{MET} < 100 \text{ GeV}$	22654	0.132004	0.129924	0.129095	0.130467
$VH_{b\bar{b}}$	23686	0.132823	0.129032	0.131202	0.131303
$VH_{b\bar{b}} + \text{MET} > 20 \text{ GeV}$	17512	0.135974	0.13137	0.13366	0.132341
$VH_{b\bar{b}} + \text{MET} > 40 \text{ GeV}$	8636	0.140116	0.135415	0.140013	0.140551
$VH_{b\bar{b}} + \text{MET} > 70 \text{ GeV}$	2592	0.143505	0.15228	0.151469	0.155914

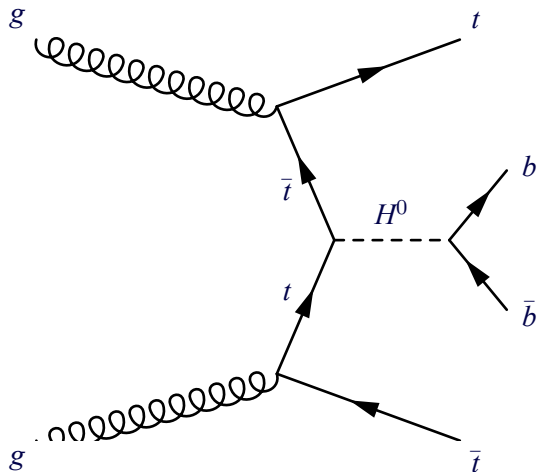
$m_{b\bar{b}}$ resolution results (Gaussian fit) for training with epochs of interest:

		Epochs			
		40	100	145	300
Subset	Training	0.137	0.138	0.138	0.138
	Training test	0.139	0.139	0.139	0.139

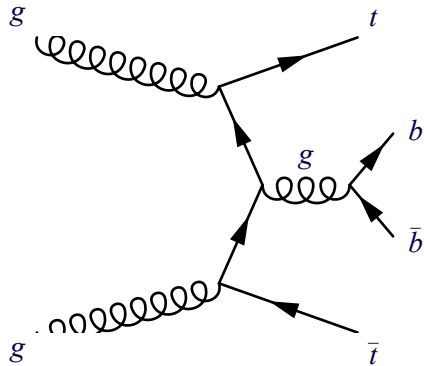
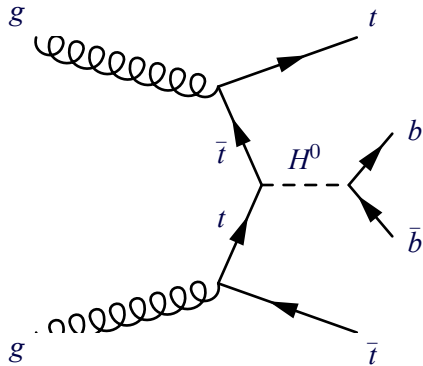
Comparison of $m_{b\bar{b}}$ resolutions for various channels both excluding and including the MET variable with various epochs:

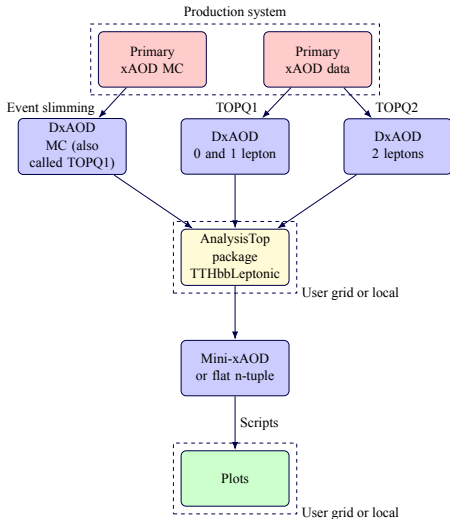
Number of epochs		$l\nu bb$	$llbb$	$\nu\nu bb$	all
50	Without MET	0.135159	0.138616	0.135159	0.137488
	With MET	0.130047	0.137266	0.136842	0.138516
	Change	-3.78%	-0.97%	+1.24%	+0.75%
100	Without MET	0.134537	0.138781	0.13656	0.13743
	With MET	0.129719	0.137265	0.136247	0.138948
	Change	-3.58%	-1.09%	-0.22%	+1.1%
150	Without MET	0.13676	0.138464	0.137943	0.13747
	With MET	0.138292	0.137261	0.137344	0.138948
	Change	+1.12%	-0.87%	-0.43%	+1.07%
500	Without MET	0.139041	0.139451	0.13849	0.13827
	With MET	0.139225	0.137261	0.136398	0.138948
	Change	+0.13%	+1.6%	-1.51%	-0.48%





Nulla consequat massa quis enim. Donec pede justo, fringilla vel, aliquet nec, vulputate eget, arcu. In enim justo, rhoncus ut, imperdiet a, venenatis vitae, justo. Nullam dictum felis eu pede mollis pretium.





Fim